

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

ACP 314 — Past Questions & Materials
by Renance

Course: ACP 314
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 300

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Paper 1 of 2 — ACP 314 SEISMIC EXPLORATION METHOD 1 2019/2020

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF EARTH SCIENCES

FIRST SEMESTER B.Sc. (GEOPHYSICS) DEGREE EXAMINATION 2019/2020 SESSION

COURSE: AGP 314 (3 CREDITS) SEISMIC EXPLORATION METHOD I TIME ALLOWED: 3 HRS.

INSTRUCTION: ANSWER ANY THREE (3) QUESTIONS ONLY

Q1 a). Proof that;

$$a) T_{(x)} = \frac{\sqrt{1 + Z^2} - X^2}{V} \quad (15 \text{ marks})$$

ai) Derive the Normal Move out correction equation (10 marks)

Q2. With a diagram explain dispersion of Wave at an interface (5 marks)

a) Sketch and explain Basic Exploration workflow (7 marks)

b) Define surface wave. With a diagram describe Rayleigh Waves and Love Waves (5 marks)

c) With a sketch explain stress and strain. (4 marks)

d) With a sketch differentiate between seismic refraction and reflection (4 marks)

Q3 a). Define the terms (i) **reflection coefficient** and (ii) **transmission coefficient** (6 marks)

b). What is the significance of the reflection coefficient being **unity** at a seismic boundary? (4 marks)

c). Consider a **600 m** thick layer of sandstone of velocity **2,500 m/s** and density **2,700 kg/m³** overlying a granite basement of velocity **3,700 m/s** and density of **2,800 kg.m³**. A seismic wave of amplitude **A = 1.0** is incident normally at the interface between the two layers. Assuming no amplitude loss due to geometrical spreading, attenuation and scattering, determine:

i). the amplitude of the wave **reflected** from the interface; — **0.21** (4 marks)

ii) the amplitude of the wave **transmitted** through the interface — **0.79** (4 marks)

iii). the **arrival time** of the reflected wave at the surface. — **0.48** (4 marks)

Sketch the seismogram recorded at the detector based on your computation. (3 marks)

Q4 a). i. Sketch the shot gather shown and on it, clearly label the **direct P-wave**, **ground roll** and the **reflected P-wave**. (9 marks)

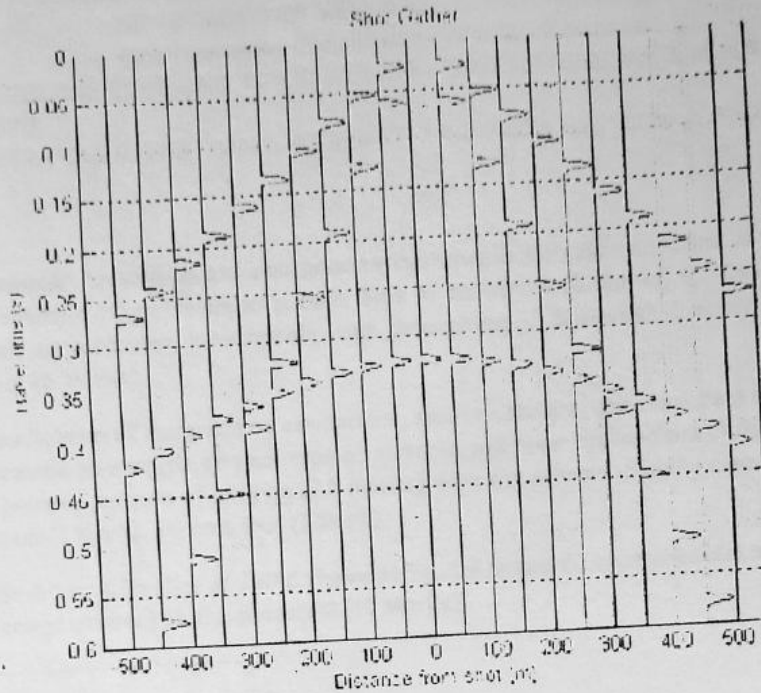
ii). What is the approximate **zero offset time** of the reflection? (3 marks)

b). Distinguish between **moveout** and **normal moveout**. (6 marks)

c). From the shot gather, what is the approximate:

a). **normal moveout** for the receiver located at **400 m** from the shot? (3 marks)

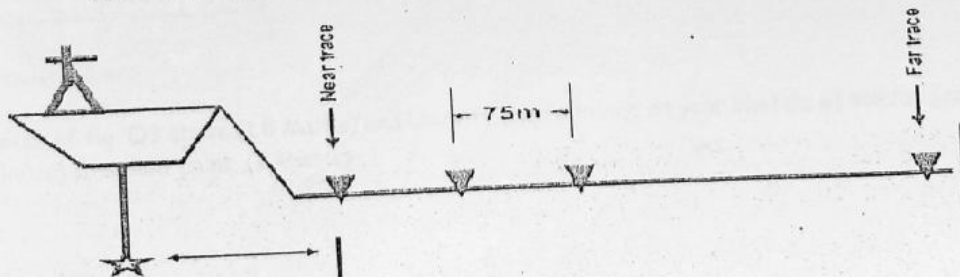
b). **moveout** between the reflection at the **100 m** and **500 m** offsets? (4 marks)



Q5 a). What is multiplicity? (5 marks)

b). The diagram below shows a seismic data acquisition setup in a marine environment. The Near and Far trace represent the first and last Receiver stations, the last station being the 480th station.

Given that the Near trace offset is 100 m, Group interval is 75 m and Shot point interval is 25 m, determine the following:



- | | | | |
|-------|-------------------------------------|------------|-----------|
| (i) | total active streamer length | = 35,925m | (4 marks) |
| (ii) | far trace offset | = 36,025m | (4 marks) |
| (iii) | offset to receiver group 300 | = 22,525 | (4 marks) |
| (iv) | offset of CMP to receiver group 300 | = 11,312.5 | (4 marks) |
| (v) | fold of the data | = 720 | |

Paper 2 of 2 — ACP 314 SEISMIC EXPLORATION METHOD 1 2019,2020

Level 300 • First Semester • PastQuestion

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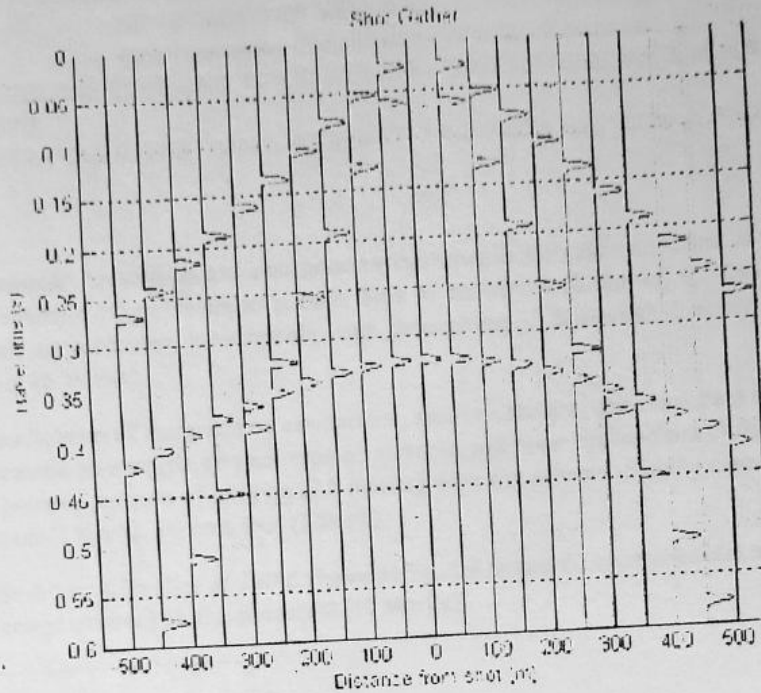
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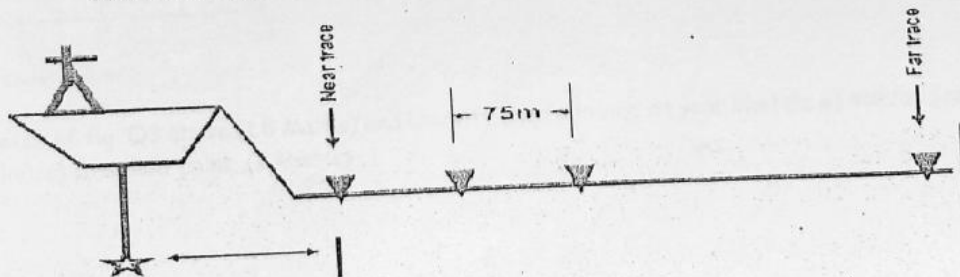
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